

Ê PHASOR TEMPLATES (Air: ε<sub>r</sub>=μ<sub>r</sub>=1)

|                                                                                                                   |              |              |              |
|-------------------------------------------------------------------------------------------------------------------|--------------|--------------|--------------|
| Wavenumber per medium <i>nonmag.</i>                                                                              |              |              |              |
| $k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$ |              |              |              |
| Direction table $u \in \{x, y, z\}$ = axis perp. to boundary                                                      |              |              |              |
| Inc dir                                                                                                           | Inc          | Refl         | Trans        |
| +û                                                                                                                | $e^{-jk_1u}$ | $e^{+jk_1u}$ | $e^{-jk_2u}$ |
| -û                                                                                                                | $e^{+jk_1u}$ | $e^{-jk_1u}$ | $e^{+jk_2u}$ |

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| General template (incident has pol. ê, amplitude a <sub>0</sub> ):                        |                           |
| $\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)                                | (incident wave, Med 1)    |
| $\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)                         | (reflected wave, Med 1)   |
| $\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)                           | (transmitted wave, Med 2) |
| (lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$ )   |                           |
| Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$ . |                           |
| ê by polarization plug into templates above                                               |                           |

| Pol.        | ê                                                    | Note               |
|-------------|------------------------------------------------------|--------------------|
| Lin. x̂     | x̂                                                   | E <sub>y</sub> = 0 |
| Lin. ŷ      | ŷ                                                    | E <sub>x</sub> = 0 |
| Lin. at 45° | (x̂ + ŷ)/√2                                          | in phase           |
| RHC         | x̂ - jŷ                                              | δ = -π/2           |
| LHC         | x̂ + jŷ                                              | δ = +π/2           |
| General     | a <sub>x</sub> x̂ + a <sub>y</sub> e <sup>jδ</sup> ŷ | elliptical         |

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

|                                                                     |                                           |                                                                                                   |                                                                                                   |
|---------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <i>Med. 1 (z &lt; 0) incident side; Med. 2 (z ≥ 0) transmitted.</i> |                                           |                                                                                                   |                                                                                                   |
|                                                                     | ⊥ pol<br>(θ <sub>i</sub> = 0)             | pol                                                                                               |                                                                                                   |
| Γ refl. wav                                                         | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$ | $\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$ |
| τ trans. wav                                                        | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$                       | $\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$                       |
| τ rel.                                                              | τ = 1 + Γ                                 | τ <sub>⊥</sub> = 1 + Γ <sub>⊥</sub>                                                               | τ <sub>  </sub> = (1 + Γ <sub>  </sub> ) $\frac{\cos \theta_i}{\cos \theta_t}$                    |
| R refl. pwr                                                         | Γ  <sup>2</sup>                           | Γ <sub>⊥</sub>   <sup>2</sup>                                                                     | Γ <sub>  </sub>   <sup>2</sup>                                                                    |
| T trans. pwr                                                        | τ  <sup>2</sup> $\frac{\eta_1}{\eta_2}$   | τ <sub>⊥</sub>   <sup>2</sup> $\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$                 | τ <sub>  </sub>   <sup>2</sup> $\frac{2 \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$              |
| R + T total                                                         | T = 1 - R                                 | T <sub>⊥</sub> = 1 - R <sub>⊥</sub>                                                               | T <sub>  </sub> = 1 - R <sub>  </sub>                                                             |

Notes: sin θ<sub>t</sub> = √μ<sub>1</sub>ε<sub>1</sub>/μ<sub>2</sub>ε<sub>2</sub> sin θ<sub>i</sub>; nonmag. ⇒ η<sub>2</sub>/η<sub>1</sub> = n<sub>1</sub>/n<sub>2</sub>.

|                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intrinsic impedance η <sub>i</sub> plug into table above                                                                                                                  |
| $\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$ $\eta_0 = 120\pi \approx 377 \Omega$ |
| Default μ <sub>r</sub> = 1 (air, dielectric). Use μ <sub>r</sub> ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.                                               |
| Low-loss correction η <sub>c</sub> σ/(ωε) ≪ 1                                                                                                                             |
| $\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$             |
| For Γ/τ just use η <sub>0</sub> /√ε <sub>r</sub> . The j term is for  η <sub>c</sub>  , θ <sub>η</sub> . See LOSSY MEDIA — 5 CASES.                                       |

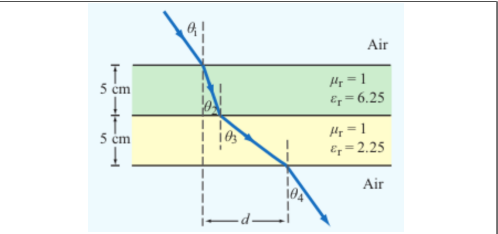
POWER REFLECTION & TRANSMISSION

|                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|
| % reflected <i>always</i> %P <sub>r</sub> = 100  Γ  <sup>2</sup>                                            |
| % transmitted <i>η-ratio matters!</i> %P <sub>t</sub> = 100  τ  <sup>2</sup> η <sub>1</sub> /η <sub>2</sub> |
| Check: %P <sub>r</sub> + %P <sub>t</sub> = 100.                                                             |

SNELL'S LAW — OBLIQUE INCIDENCE

|                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------|
| Single boundary <i>angles from normal</i>                                                                                |
| $\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$               |
| Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4                                                                     |
| $\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$                                       |
| Air-slabs-air shortcut <i>outer media identical</i>                                                                      |
| θ <sub>exit</sub> = θ <sub>incident</sub>                                                                                |
| <i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i> |

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t<sub>i</sub>, refracted angle inside slab i Slab-indexed θ<sub>i</sub> = angle inside slab i; t<sub>i</sub> = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ<sub>i</sub>) in the formula is always the one the ray makes inside that slab of height (t<sub>i</sub>).

LOSSY MEDIA — 5 CASES

| Loss tangent ε' = ε <sub>r</sub> ε <sub>0</sub> ; ε <sub>0</sub> = 8.854 × 10 <sup>-12</sup> F/m |                                                                              |                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                  | $\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$ |                                                                                                                                                                                                                 |
| Type                                                                                             | σ/ωε                                                                         | η <sub>c</sub> , α, β                                                                                                                                                                                           |
| Lossless (σ = 0)                                                                                 | 0                                                                            | η = η <sub>0</sub> /√ε <sub>r</sub> exact<br>α = 0, β = ω√ε <sub>r</sub> /c                                                                                                                                     |
| Low-loss                                                                                         | < 10 <sup>-2</sup>                                                           | η <sub>c</sub> ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon'}{2\omega \epsilon_r}\right)$<br>α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ , β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.                                                                                      | 10 <sup>-2</sup> –10 <sup>2</sup>                                            | exact: (η/√X)e <sup>jθ<sub>η</sub></sup><br>see below                                                                                                                                                           |
| Good cond.                                                                                       | > 10 <sup>2</sup>                                                            | η <sub>c</sub> ≈ (1 + j)√πfμσ<br>α ≈ β ≈ √πfμσ                                                                                                                                                                  |
| Magnetic                                                                                         | any                                                                          | keep full √μ <sub>r</sub> /ε <sub>r</sub> η <sub>0</sub>                                                                                                                                                        |

For Γ/τ: drop j correction; use η<sub>0</sub>/√ε<sub>r</sub>. Magnetic: keep μ<sub>r</sub>.

|                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-loss quick params σ/(ωε) ≪ 1, μ <sub>r</sub> = 1                                                                                                             |
| α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), η <sub>c</sub> ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω) |
| Exact (quasi-cond) X = √[1 + (ε''/ε') <sup>2</sup> ]                                                                                                             |
| α = ω√ $\frac{\mu \epsilon'}{2}$ √X-1, β = ω√ $\frac{\mu \epsilon'}{2}$ √X+1                                                                                     |
| θ <sub>η</sub> = $\frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$                                                                                           |

RECTANGULAR WAVEGUIDE — MODES

|                                                                                                            |                                                     |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| TE mode <i>E<sub>z</sub> = 0, H<sub>z</sub> ≠ 0</i>                                                        | TM mode <i>H<sub>z</sub> = 0, E<sub>z</sub> ≠ 0</i> |
| Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.                                                       |                                                     |
| E <sub>x</sub> form (TE) read m, n from arguments                                                          |                                                     |
| $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$ |                                                     |
| coef. on x = mπ/a; coef. on y = nπ/b.                                                                      |                                                     |

Identify TE<sub>mn</sub> vs TM<sub>mn</sub> cos/sin pattern of E<sub>x</sub> is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H<sub>z</sub>(x, y) given ⇒ TE. Source field E<sub>z</sub>(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE<sub>10</sub> exists but TM<sub>10</sub> doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e<sup>-jβz</sup> factor omitted. k<sub>c</sub><sup>2</sup> = (mπ/a)<sup>2</sup> + (nπ/b)<sup>2</sup>. Ê in V/m, Ĥ in A/m, Z in Ω.

|                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------|
| TE mode <i>generator: Ê<sub>z</sub></i>                                                                               |
| $\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$                                                       |
| $\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$  |
| $\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$ |
| $\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$                                |
| $Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$                                                                                |

|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| TM mode <i>generator: Ê<sub>z</sub></i>                                                                           |
| $\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$                                                   |
| $\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$ |
| $\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$ |
| $\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$                            |
| $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$                                                                              |

TEM plane wave *for reference*

$$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$$
$$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega\sqrt{\mu\epsilon}$$

|                                                                          |  |
|--------------------------------------------------------------------------|--|
| Properties common to TE/TM                                               |  |
| $f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2}$ (Hz)                   |  |
| $\beta = k\sqrt{1 - (f_c/f)^2}$ (rad/m)                                  |  |
| $u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}}$ (m/s) |  |

CUTOFF FREQUENCY

| Common cutoffs (a > b) f <sub>c</sub> formula in Table 8-3                                                                                                |                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Mode                                                                                                                                                      | f <sub>mn</sub>                                           |
| u <sub>p0</sub>                                                                                                                                           | c/√ε <sub>r</sub> (hollow: u <sub>p0</sub> = c)           |
| TE <sub>10</sub>                                                                                                                                          | f <sub>10</sub> = u <sub>p0</sub> /(2a) (dominant)        |
| TE <sub>01</sub>                                                                                                                                          | f <sub>01</sub> = u <sub>p0</sub> /(2b)                   |
| TE <sub>20</sub>                                                                                                                                          | f <sub>20</sub> = u <sub>p0</sub> /a = 2f <sub>10</sub>   |
| TE <sub>11</sub> , TM <sub>11</sub>                                                                                                                       | f <sub>11</sub> = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$ |
| Propagation requires f > f <sub>mn</sub> . In Z <sub>TE</sub> /Z <sub>TM</sub> and u <sub>g</sub> , f <sub>c</sub> = f <sub>mn</sub> of the excited mode. |                                                           |

ε<sub>r</sub> FROM DISPERSION

|                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|
| Given ω, β, mode (m, n) <i>result is unitless</i>                                                                                     |
| $\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless) |
| Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 <sup>8</sup> (m/s), m, n (unitless).                                               |

GROUP VELOCITY & TRAVEL TIME

|                                                                    |
|--------------------------------------------------------------------|
| $u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s), $u_p u_g = u_{p0}^2$ |
| t = L/u <sub>g</sub> (s) travel time through guide of length L     |
| Higher modes have lower u <sub>g</sub> ⇒ arrive later.             |

BOUNDARY

|                                                      |
|------------------------------------------------------|
| η <sub>i</sub> , η <sub>t</sub> — imp. (Ω)           |
| Γ, τ, R, T — coeffs (unitless)                       |
| τ = 1 + Γ; R =  Γ  <sup>2</sup>                      |
| k <sub>i</sub> = ω√ε <sub>r,i</sub> /c (rad/m)       |
| α <sub>2</sub> (Np/m), β <sub>2</sub> (rad/m): lossy |
| z = 0 — boundary plane (m)                           |
| η <sub>0</sub> = 120π ≈ 377 Ω                        |

SNELL / OBLIQUE

|                                                      |
|------------------------------------------------------|
| θ <sub>i</sub> , θ <sub>t</sub> — angles from normal |
| n <sub>i</sub> = √ε <sub>r,i</sub> — index of refr.  |
| t <sub>i</sub> — slab thickness (m)                  |
| d — lateral disp. (m)                                |
| plane of inc. = plane of k+normal                    |
| ⊥: E ⊥ this plane                                    |
| : E    this plane                                    |

WAVEGUIDE

|                                      |
|--------------------------------------|
| a, b — wide, narrow dim. (m);        |
| a > b                                |
| m, n — mode indices (int.)           |
| f <sub>mn</sub> — cutoff (Hz)        |
| β — prop. const (rad/m)              |
| Z <sub>TE</sub> /TM — wave imp. (Ω)  |
| E <sub>x</sub> — E-field comp. (V/m) |
| H <sub>y</sub> — H-field comp. (A/m) |
| η = η <sub>0</sub> /√ε <sub>r</sub>  |

VELOCITIES

|                                                                            |
|----------------------------------------------------------------------------|
| u <sub>p0</sub> = c/√ε <sub>r</sub> — bulk phase vel.                      |
| u <sub>p</sub> = u <sub>p0</sub> /√[1 - (f <sub>c</sub> /f) <sup>2</sup> ] |
| u <sub>g</sub> = u <sub>p0</sub> √[1 - (f <sub>c</sub> /f) <sup>2</sup> ]  |
| u <sub>p</sub> u <sub>g</sub> = u <sub>p0</sub> <sup>2</sup>               |
| t = L/u <sub>g</sub> — travel time                                         |
| c = 3 × 10 <sup>8</sup> m/s                                                |

POWER & UNITS

|                                                                      |
|----------------------------------------------------------------------|
| %P <sub>r</sub> , %P <sub>t</sub> — pwr refl./trans. (unitless)      |
| %P <sub>r</sub> = 100 Γ  <sup>2</sup>                                |
| %P <sub>t</sub> = 100 τ  <sup>2</sup> η <sub>1</sub> /η <sub>2</sub> |
| %P <sub>r</sub> + %P <sub>t</sub> = 100                              |
| σ (S/m), ε (F/m), μ (H/m)                                            |
| ε <sub>0</sub> = 8.85 × 10 <sup>-12</sup> F/m                        |
| μ <sub>0</sub> = 4π × 10 <sup>-7</sup> H/m                           |

## DIPOLE TYPE BY LENGTH

**STEP 0 — always start here**  $\lambda = \text{wavelength (m), } l = \text{antenna rod length (m), } c = 3 \times 10^8 \text{ m/s}$

| $\lambda = \frac{c}{f} \text{ (m)}$ | ratio $\frac{l}{\lambda} \rightarrow$ | pick row below                                                |
|-------------------------------------|---------------------------------------|---------------------------------------------------------------|
| $l/\lambda$                         | Type                                  | Use formula                                                   |
| $< 1/50$                            | Hertzian (short)                      | HERTZIAN $S(R, \theta)$ , $D = 1.5$                           |
| $= 1/4$                             | Quarter-wave                          | ARB formula, $\pi l/\lambda = \pi/4$                          |
| $= 1/2$                             | Half-wave                             | half-wave shortcut, $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ |
| $= 1$                               | Full-wave                             | ARB, $D = 2.41$ , $R_{\text{rad}} = 199 \Omega$               |
| other                               | Arbitrary                             | ARB formula                                                   |

If  $l/\lambda < 1/50$  STOP and use Hertzian. Do NOT plug small  $l$  into the arbitrary formula — wrong shape.

## HERTZIAN DIPOLE — $S(R, \theta)$

|                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Far-field E,H phasors <small>small <math>l \ll \lambda</math></small>                                                                                         |
| $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$ , $\vec{H}_\phi = \vec{E}_\theta / \eta_0$                                             |
| Power density $l/\lambda < 1/50$ ; far field                                                                                                                  |
| $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$                                 |
| $l$ rod length (m); $I_0$ peak current (A); $R$ obs. dist. (m); $\theta$ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$ . |
| Max @ broadside $\theta_{\text{max}} = \pi/2$ ; $D$ exact $D = 1.5$ (1.76 dB)                                                                                 |
| Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$                                                                    |

## ARBITRARY-LENGTH DIPOLE — $S(\theta)$

|                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power density at $R$ any $l$                                                                                                                                                             |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$                                        |
| At $\theta = 0, \pi$ : bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$ . |
| Normalized pattern <small>dimensionless</small> $F(\theta) = S(\theta)/S_{\text{max}}$                                                                                                   |
| Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$ , $\cos(\pi/4) = \sqrt{2}/2$                                                                                                        |
| $\frac{S(\theta)}{S_0} = \left[ \frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$ , $S_0 = \frac{15 I_0^2}{\pi R^2}$                                   |
| $S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$                                                                                               |
| $D = 1.64$ , $R_{\text{rad}} = 36.5 \Omega$ , $\theta_{\text{max}} = \pi/2$ .                                                                                                            |
| Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$ , <small>shortcut form</small>                                                                                                         |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$                                                                                           |
| $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ , $\theta_{\text{max}} = \pi/2$ .                                                                                                              |

## COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

|                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|
| Conversion <small>multiply by <math>\pi/180</math> or <math>180/\pi</math></small>                                                  |
| $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$ |
| $^\circ$ rad    sin $\theta$ cos $\theta$ sin <sup>2</sup> $\theta$ cos <sup>2</sup> $\theta$                                       |
| 0    0    0    1    0    1                                                                                                          |
| 30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4    3/4                                                                                              |
| 45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2    1/2                                                                                     |
| 60 $\pi/3$ $\sqrt{3}/2$ 1/2    3/4    1/4                                                                                           |
| 90 $\pi/2$ 1    0    1    0                                                                                                         |
| 180 $\pi$ 0    -1    0    1                                                                                                         |

Antenna formulas usually take  $\theta$  in rad. RAD mode on calc.

## LINEAR ANTENNA ARRAY

|                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| Setup $N$ identical elements along axis, spacing $d$                                                                       |
| elem pos $z_i = id$ , $i = 0, \dots, N-1$ ; $A_i = a_i e^{j\psi_i}$                                                        |
| Equal phase ( $\psi_i = 0$ ) $\Rightarrow$ broadside beam ( $\theta_{\text{max}} = \pi/2$ ).                               |
| Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$ .               |
| Pattern multiplication <small><math>S_e =</math> single element pattern</small>                                            |
| $S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$                                                                    |
| Array factor (general) $k = 2\pi/\lambda$                                                                                  |
| $F_a(\theta) = \left  \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$                                                 |
| Uniform $A_i = 1$ , equal phase <small>closed form</small>                                                                 |
| $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$ , $\gamma = kd \cos \theta$                                     |
| Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$ ). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$ . |
| HPBW (broadside, uniform) <small>use <math>Nd</math>, not <math>(N-1)d</math></small>                                      |
| $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) $= 50.8^\circ \cdot \frac{\lambda}{Nd}$                                      |
| Grating lobes appear if $d > \lambda$ (broadside). Avoid.                                                                  |

## DIPOLE

|                                                  |
|--------------------------------------------------|
| $l$ — antenna length (m)                         |
| $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s     |
| $I_0$ — peak current (A)                         |
| $k = 2\pi/\lambda$ (rad/m)                       |
| $R$ — obs. dist. (m); $\theta$ from axis         |
| $\eta_0 = 120\pi \approx 377 \Omega$             |
| $S(R, \theta)$ — pwr density (W/m <sup>2</sup> ) |

## BEAM

|                                                                |
|----------------------------------------------------------------|
| $F(\theta) = S/S_{\text{max}}$ (unitless)                      |
| $a$ — coef. on $\theta^2$ in Gaussian                          |
| $I_0$ — HPBW (rad or $^\circ$ )                                |
| $\Omega_p$ — pattern solid angle (sr)                          |
| $D = 4\pi/\Omega_p$ — directivity                              |
| $\xi$ — rad. efficiency; $G = \xi D$                           |
| $D_{\text{dB}} = 10 \log_{10} D$ ; $D = 10^{D_{\text{dB}}/10}$ |

## FRIIS / dB

|                                                  |
|--------------------------------------------------|
| $P_t$ — power transmitted (W)                    |
| $P_r$ — power received (W)                       |
| $G_t$ — transmit gain (linear)                   |
| $G_r$ — receive gain (linear)                    |
| $S_r = G_t P_t / (4\pi R^2)$ (W/m <sup>2</sup> ) |
| $P_r = G_t G_r (\lambda / 4\pi R)^2 P_t$         |
| $G = 10^{G_{\text{dB}}/10}$                      |
| 3 dB = $\times 2$ , 10 dB = $\times 10$          |
| $P_N = k_B T_{\text{sys}} B$ (W)                 |

## APERTURE

|                                                |
|------------------------------------------------|
| $A_p$ — physical area (m <sup>2</sup> )        |
| $A_e = \lambda^2 D / (4\pi)$ (m <sup>2</sup> ) |
| $A_e \approx A_p$ for aperture                 |
| $D \approx 4\pi A_p / \lambda^2$ (unitless)    |
| $\beta \approx 0.88 \lambda / \ell$ (rect/sq.) |
| $\beta \approx 1.02 \lambda / d$ (circular)    |
| $2A_p \Rightarrow 2D$ , $\beta/\sqrt{2}$       |

## ARRAY

|                                                 |
|-------------------------------------------------|
| $N$ — number of elements                        |
| $d$ — spacing (m)                               |
| $\gamma = kd \cos \theta$                       |
| $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$    |
| $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$    |
| $F_a^{\text{norm}} = F_a/N^2$                   |
| $\beta \approx 0.88 \lambda / (Nd)$ (broadside) |

## BEAM PARAMETERS — $\beta$ , $\Omega_p$ , $D$

|                                                                                |
|--------------------------------------------------------------------------------|
| Normalize <small>always start here</small>                                     |
| $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)                              |
| Beamwidth $\beta$ at threshold $X$                                             |
| Set $F(\theta_X) = X$ , solve for $\theta_X$ ; $\beta = 2\theta_X$ (symmetric) |
| Threshold $X = 10^{\text{dB}/10}$ $\theta_X$ (Gauss. short-cut)                |
| HPBW (−3 dB)    0.5 $\sqrt{\frac{\ln 2}{a}}$                                   |
| −6 dB    0.25 $\sqrt{\frac{\ln 4}{a}}$                                         |
| −10 dB    0.1 $\sqrt{\frac{\ln 10}{a}}$                                        |
| any $F(\theta_X)$ any $X$ $F(\theta_X) = X$                                    |

|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| Pattern solid angle $\Omega_p$ <small>recognize pattern, look up</small>                                                                  |
| $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)                                                            |
| No $\phi$ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$ .                                                              |
| $\theta$ is the integration variable — DON'T plug in a number for $\theta$ (e.g. $\theta_{1/2}$ ). Integrate over $\theta \in [0, \pi]$ . |

| Pattern $F(\theta)$           | $\Omega_p$ (sr)                     |
|-------------------------------|-------------------------------------|
| Isotropic ( $F=1$ )           | $4\pi \approx 12.57$                |
| Hertzian $\sin^2 \theta$      | $8\pi/3 \approx 8.38$               |
| Half-wave dipole              | $\approx 7.66$                      |
| Narrow Gauss $e^{-a\theta^2}$ | $\pi/a = \pi\theta_{1/2}^2 / \ln 2$ |
| 2-axis HPBW (aperture)        | $\beta_{xz} \cdot \beta_{yz}$       |
| Other (use integral)          | numerical                           |

TI-36X Pro: [2nd] [d/d $\Delta$ ] for  $\int_{\square}^{\square}$ , RAD mode, multiply by 2 $\pi$ .

|                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Directivity $D$ <small>always <math>D = 4\pi/\Omega_p</math>; common shortcuts below</small>                                                                                                    |
| $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$                                                                                                                         |
| WORD TRAP: “direction of max radiation” = angle $\theta_{\text{max}}$ (rad/deg). “directivity” = unitless number $D$ . Different things!                                                        |
| Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$ . 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$ . Circular: $D \approx 4\pi / \beta^2$ . |
| Gain (if asked) $\xi = \text{rad. eff.}$ , $0 < \xi \leq 1$                                                                                                                                     |
| $G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$                                                                                                                                                    |
| Lossless / ideal antenna $\Rightarrow G = D$ .                                                                                                                                                  |

## COMMON DIPOLE PARAMETERS

| Type      | $l$               | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}}$ ( $\Omega$ ) |
|-----------|-------------------|------|-----------------|-------------------------------|
| Isotropic | N/A               | 1    | 0               | —                             |
| Hertzian  | $< \lambda/50$    | 1.5  | 1.76            | $80\pi^2 (l/\lambda)^2$       |
| Half-wave | $\lambda/2$       | 1.64 | 2.15            | 73.2                          |
| Monopole  | $\lambda/4$ (gnd) | 1.64 | 2.15            | 36.6                          |
| Full-wave | $\lambda$         | 2.41 | 3.82            | 199                           |

Half-wave:  $P_{\text{rad}} = 36.6 I_0^2 \text{ W}$ . Monopole ( $\lambda/4$  above ground): same patt. as half-wave but  $P_{\text{rad}}$ ,  $R_{\text{rad}}$  halved. Lossless antenna ( $\xi = 1$ )  $\Rightarrow G = D$ . So in Friis, use  $G_t$ ,  $G_r =$  these  $D$  values (e.g. half-wave  $G = 1.64$ ).

## ANTENNA AREAS ( $A_e$ , $A_p$ )

|                                                                                                            |
|------------------------------------------------------------------------------------------------------------|
| Effective area <small>antenna's RX cross section; <math>D</math> from BEAM PARAMS or table above</small>   |
| $A_e = \frac{\lambda^2 D}{4\pi}$ (m <sup>2</sup> )                                                         |
| Physical cross section <small>wire dipole, diameter <math>d</math>; <math>l</math> see table above</small> |
| $A_p = l d$ (m <sup>2</sup> , any dipole)                                                                  |
| Inverse: $D$ from $A_e$                                                                                    |
| $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)                                                                |
| Hertzian special case $D = 1.5$                                                                            |
| $A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2$ (m <sup>2</sup> )                   |
| Thin wire: $A_e \gg A_p$ . Antenna captures field over $\sim \lambda$ .                                    |

## FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr  $G_t$ ,  $G_r$  linear gains;  $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent:  $P_r = S_r A_{e,r}$ .

Solve for  $R$  max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using  $A_e$  recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss:  $L_{fs} = (4\pi R/\lambda)^2$ , so  $P_r = G_t G_r P_t / L_{fs}$ .

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

$F_t, F_r$  normalized patterns at the link directions;  $F = 1$  when aligned.

Recipe 1.  $\lambda = c/f$ . 2. Convert dB gains to linear:  $G = 10^{G_{\text{dB}}/10}$ .

3. If TX is “half-wave dipole” use  $G_t = 1.64$ . 4. Plug into Friis.

Use linear  $G$ , not dB. Convert FIRST.

## dB $\leftrightarrow$ LINEAR

Works for any power-like quantity:  $G$  gain,  $D$  directivity,  $P_r/P_t$ , SNR, etc.

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g.  $D_{\text{dB}} = 10 \log_{10} D$ ;  $G_{\text{dB}} = 10 \log_{10} G$ ;  
 $SNR_{\text{dB}} = 10 \log_{10} (P_r/P_N)$ .

| dB | linear | dB  | linear |
|----|--------|-----|--------|
| 0  | 1      | 10  | 10     |
| 3  | 2      | 13  | 20     |
| 6  | 4      | 20  | 100    |
| 7  | 5      | 30  | 1000   |
| −3 | 0.5    | −10 | 0.1    |

Memorize: 3 dB  $\Leftrightarrow \times 2$ , 10 dB  $\Leftrightarrow \times 10$ . Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power  $\propto E^2$ ).

## NOISE & SNR

Noise power thermal noise into matched receiver

$$P_N = k_B T_{\text{sys}} B \text{ (W)}$$

$k_B = 1.38 \times 10^{-23}$  J/K;  $T_{\text{sys}}$  system noise temp (K);  $B$  receiver bandwidth (Hz).

Signal-to-noise ratio

$$SNR = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \quad SNR_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs  $SNR > 10$  dB for usable signal.

## APERTURE ANTENNAS (horn, dish)

$A_p$  physical aperture area (m<sup>2</sup>);  $A_e$  effective area (m<sup>2</sup>,  $= \lambda^2 D / 4\pi$ );  $\beta_{xz}, \beta_{yz}$  HPBW in two planes (rad);  $\ell$ ,  $d$  aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad)}; \text{ circular: } D \approx 4\pi / \beta^2$$

HPBW and  $D$  by aperture shape uniform illum.;  $\beta$  in rad

| Shape                        | $A_p$           | HPBW (rad)                                                                         | $D$ (unitless)                                                             |
|------------------------------|-----------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Rect. $\ell_x \times \ell_y$ | $\ell_x \ell_y$ | $\beta_x \approx 0.88 \lambda / \ell_x$<br>$\beta_y \approx 0.88 \lambda / \ell_y$ | $D \approx 4\pi / (\beta_x \beta_y)$<br>$= 4\pi \ell_x \ell_y / \lambda^2$ |
| Square $\ell$                | $\ell^2$        | $\beta \approx 0.88 \lambda / \ell$                                                | $D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$           |
| Circular (diam. $d$ )        | $\pi d^2/4$     | $\beta \approx 1.02 \lambda / d$                                                   | $D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$             |

Scaling rules  $D \propto A_p / \lambda^2$ ;  $\beta \propto \lambda / \sqrt{A_p}$

| Change                                          | New $D$                            | New $\beta$                                          |
|-------------------------------------------------|------------------------------------|------------------------------------------------------|
| Double area ( $A_p \rightarrow 2A_p$ )          | $D_{\text{new}} = 2D_{\text{old}}$ | $\beta_{\text{new}} = \beta_{\text{old}} / \sqrt{2}$ |
| Double freq ( $\lambda \rightarrow \lambda/2$ ) | $D_{\text{new}} = 4D_{\text{old}}$ | $\beta_{\text{new}} = \beta_{\text{old}} / 2$        |